Digital Methods Spring 2025: syllabus and course plan

Course runs weeks 15-22, Tuesdays and Thursdays (lectures 8-10, exercise classes 10-13).

Note: As a rule, exercise classes are used for project-related work. More in-depth descriptions of each exercise class will be made available on a running basis.

Course description:

Using digital methods is a specific approach to doing digital social research. In digital methods, the focus is placed on the digital media contexts where data is generated as a by-product of social interaction, and on new ways of combining quantitative and qualitative methods of digital inquiry and analysis. This course provides students with practical skills in implementing three sets of computer-assisted qualitative methods: exploratory network analysis, digital online ethnography, and content analysis. As such, it supplements the various quantitative techniques taught in other courses on the degree programme, and provides tools for mixing qualitative methods with textual and/or visual quantitative data into qualitative-quantitative social-science analyses. Students train these skills by conducting their own integrated mapping of a public issue, involving networks, ideas, and behavior across individual and organizational levels and across multiple digital platforms.

Exam form:

Group-based oral exam with a prior written assignment in groups of 3-4 students. The written assignment should contain a description of method accounts, forms of analyses, and formulate and evaluate an explicit research question. In addition, it may contain documentation in the shape of code, field notes, data visualizations, and so on, as relevant to the project.

The total length of the written assignment must not exceed:
• For three students: 20 standard pages
• For four students: 25 standard pages
In the standard situation (a 3-person group), the oral part of the exam lasts 40 minutes in total (30 minutes of examination). With extra student in group, 5 minutes are added. Aid: All aids allowed. Marking scale: 7-point grading scale. Censorship form: No external censorship.

Intended learning outcomes:
Knowledge

- Show familiarity with the basic techniques, use scenarios, and validity criteria of computer-assisted qualitative methods, i.e. digital ethnography, content analysis, and exploratory network analysis.
- Account for the procedures, potentials, and pitfalls of combining qualitative and quantitative data sources, including in integrated quali-quantitative ways.

- Account for the relationship between digital methods' emphasis on the media contexts of digital data and the broader questions, claims and biases of social data science.

Skills

- Identify the procedures of qualitative content analysis for designing appropriate semantic categories, including for use in subsequent machine learning with quantitative text (and/or visual) data.
- Extract, and communicate patterns of networks, ideas, and behaviour characteristic of specific social settings and public issues, using the appropriate qualitative method(s).
- Combine qualitative data with a quantitative data source, thereby integrating heterogeneous digital data formats into comprehensive social analyses.

Competencies

- Evaluate and analyse a social data problem from both qualitative and quantitative perspectives, including determining when to deploy which method designs.
- Design and implement small-scale digital ethnography campaigns, along with exploratory network analysis and content analysis, to obtain insights into social networks, ideas, and behaviour at individual and organizational levels.
- Combine qualitative and quantitative sources of data, as well as forms of narration and visualization, into persuasive quali-quantitative reports on social data problems for a range of organizational use scenarios.

<u>Teaching and learning methods:</u>

Teaching combines lectures and in-class method exercises with extensive out-of-class project work together with near weekly supervision. Throughout the course, students train their qualitative digital method skills by conducting their own project (with some teacher assistance available), i.e. digitally mapping a public issue chosen by themselves (or possibly suggested by the teachers). In-class exercises give priority to providing students first-hand skills in closely combining digital data formats into combined qualitative and quantitative social analyses that mirror realistic use scenarios in a range of contexts where social data analysis is a key component.

<u>Literature</u>

The syllabus consists mainly in relevant research articles pertaining to digital methods, mixed methods and the method skills and traditions covered. The syllabus is approximately 400 pages + 80 pages of self-selected literature. Groups should choose articles or book chapters relevant for their substantive projects.

Lecture	Readings	Exercises	Teachers
8/4 Lecture 1: What is mixed digital methods?	Blok, A., Bang Carlsen, H., & Albris, K. (2022). How to Conduct Quali-Quantitative Digital Methods Research. SAGE Publications, Ltd. (10 pages) Small, Mario Luis. "How to conduct a mixed methods study: Recent trends in a rapidly growing literature." Annual review of sociology 37 (2011): 57-86. (29 pages) https://www.jstor.org/stable/41288599	1) Get into groups. 2) Topic/question brainstorm: what specific topic/question would you like to work on? 3) How would one inquire into the topic/question with the help of the mixed digital methods taught in this course? 4) Do very preliminary 'searching around' for inspiration and knowledge – gather up collectively, make preliminary list of possible topics/questions .	Hjalmar
Blok 1	**Finding a question and a data-site**		
10/4 Lecture 2: Research question and mixed research designs	Frederiksen, Morten, Jonas Toubøl, and Hjalmar Bang Carlsen. "Theory and Design" "Mixed Methods: A Primer for Sociologist." 49-73 (24 pages). English version will be published on Absalon. Grigoropoulou, N., Small, M.L. The data revolution in social science needs qualitative research. Nat Hum Behav 6, 904–906 (2022). (10 pages) https://doi.org/10.1038/s41562-022-01333-7	1) Spend the exercise class doing preliminary scouting research on the topic, organizing it as an early immersion journal (cf. Kozinets). 2) Formulate a tentative research question 3) Prototype your mixed method design. What data types do you need? What types of analysis? What is the reason for mixing data types and analysis types?	Hjalmar

22/4 Lecture 3: Find, evaluate and select digital data sources	Kozinets, Chapter 8, "Investigating: Five Steps to Social Media Data Collection." Page 211-242 (31 pages). Kozinets´, Chapter 10 "Immersing: Journalizing and Organizing a Reflective Curation". Page 277-306 (29 pages) Venturini, T. et al.. 2018. 'A reality check-list for digital methods'. New Media & Society 20(11). https://doi.org/10.1177/1461444818769236. (14 pages) Kozinets, Chapter 6, "Ethics: Procedures and Flowcharts, Updates and Rules." Page 163-187 (25 pages)	1) Scout for datasites 2) Evaluate datasites 3) Select one or more datasites to proceed with as the main datasource(s). 4) Describe relevant datasites in immersion journal 5) Update research question and design if needed.	Hjalmar
24/4 Supervision 1: Find, evaluate and select digital data sources	Find a link for booking supervision time on the absalon course page.	1) Scout for datasites 2) Evaluate datasites 3) Select one or more datasites to proceed with as the main datasource(s). 4) Describe relevant datasites in immersion journal 5) Update research question and design if needed.	Hjalmar
Milestone 1:	1) Description and justification of research topic and question 2) Description and justification for central datasite(s) (assignment 1, 1-2 pages – hand in to your TA 25/4 at 23.59)		

29/4: Lecture 4: Data Design and Data collection	Frederiksen, Morten, Jonas Toubøl, and Hjalmar Bang Carlsen. "Data Design" "Mixed Methods: A Primer for Sociologist." 75-117 (42 pages). English version will be published on Absalon. Rafail, P. (2018). Nonprobability Sampling and Twitter: Strategies for Semibounded and Bounded Populations. Social Science Computer Review, 36(2), 195–211. https://doi.org/10.1177/0894439317709431 (15 pages) Bang Carlsen, Hjalmar, Tobias Gårdhus, and Jonas Toubøl. "Ukrainian refugee solidarity mobilization online." European Societies (2023): 1-12. https://doi.org/10.1080/14616696.2023.2177704 (11 pages)	1) Designing data collection and sampling strategy 2) Initial testing of data collection feasibility Feedback on milestone 1 from TA. .	Hjalmar
1/5 Supervision 2: Data Design and Data collection		1) Data collection, data collection, data collection.	Hjalmar
Milestone 2:	1) Updated Research Question 2) Data Collection Plan 3) Preliminary Data Analysis Design (assignment 1, 1-2 pages – hand in to teacher team 2/5 at 23.59)		
6/5 Lecture 5: Computer assisted discovery and open coding	Carlsen, H. B., & Ralund, S. (2022). Computational grounded theory revisited: From computer-led to computer-assisted text analysis. Big Data & Society, 9(1). (9 pages) https://doi.org/10.1177/20539517221080146 Chapter 5, "Open Coding" In Timmermans, Stefan, and Iddo Tavory. Data analysis in qualitative research: Theorizing with abductive analysis. University of Chicago Press, 2022. 69-90. (21 pages)	1) Work through an open coding session collectively in class. 2) Work on open coding of own data. 3) Data collection continued 4) Set up code for computer assisted discovery a) Run exploratory quantitative text analysis	Hjalmar

	Karell, D., & Freedman, M. (2019). Rhetorics of Radicalism. American Sociological Review, 84(4), 726-753. https://doi.org/10.1177/0003122419859519 (27 pages) *Recommended/optional:* Evans, James A., and Pedro Aceves. "Machine translation: Mining text for social theory." Annual review of sociology 42 (2016): 21-50. https://doi.org/10.1146/annurev-soc-081715-074206 Chapter 2 in Grimmer, Justin, Margaret E. Roberts, and Brandon M. Stewart. Text as data: A new framework for machine learning and the social sciences. Princeton University Press, 2022. Tavory, Iddo, and Stefan Timmermans. Abductive analysis: Theorizing qualitative research. University of Chicago Press, 2014.	b) Device effective sampling procedure for qualitative open coding Feedback on milestone 2 from TA.	
8/5 Supervision 3: Computer assisted discovery and open coding		1) Data collection continued 2) Set up code for computer assisted discovery 3) Continue with open coding and exploratory text analysis	Hjalmar
13/5 Lecture 6: Computer assisted focused coding and text classification.	Chapter 6, "Focused Coding" In Timmermans, Stefan, and Iddo Tavory. Data analysis in qualitative research: Theorizing with abductive analysis. University of Chicago Press, 2022. 91-115. (24 pages). Uploaded on absalon. MacQueen, Kathleen M., et al. "Codebook development for team-based qualitative analysis." CAM Journal10.2 (1998): 31-36. (5 pages). Uploaded on absalon. Törnberg, Petter. "Best Practices for Text Annotation with Large Language Models." Sociologica 18.2 (2024). https://sociologica.unibo.it/article/view/19461/18663 Carlsen, Hjalmar Bang, and Jonas Toubøl. "Some refugees welcome? Frame disputes between micro-cohorts in the refugee solidarity movement." (2025). Working paper. Uploaded on absalon.	1) Work through a focused coding session collectively in class. 2) Work on a closed coding session. 3) Choose quant text measure 4) Develop codebook 5) Instruct LLM for text classification OR develop dictionary OR supervised machine learning model	Hjalmar

15/5 supervision 4: Computer assisted focused coding and text classification.		1) Work on a closed coding and qual analysis 2) Choose quant text measure 3) Develop codebook 4) Instruct LLM for text classification OR develop dictionary OR supervised machine learning model	Hjalmar
Milestone 3:	1) Data Collection Update 2) Data Analysis Design Update 3) Initial analysis(mainly qualitative) Assignment 3 – hand in 16/5 at 23.59 2 pages.		
20/5 Lecture 7: Text classification (continued) and Quantitative (text) analysis	Nelson, L. K. (2021). Cycles of Conflict, a Century of Continuity: The Impact of Persistent Place-Based Political Logics on Social Movement Strategy. American Journal of Sociology, 127(1), 1–59. https://doi.org/10.1086/714915 Greve, H. R., Rao, H., Vicinanza, P., & Zhou, E. Y. (2022). Online Conspiracy Groups: Micro-Bloggers, Bots, and Coronavirus Conspiracy Talk on Twitter. American Sociological Review, 87(6), 919–949. https://doi.org/10.1177/00031224221125937	1) Work on quantitative text analysis 2) Label test set for evaluating classifier Feedback on milestone 3 from TA.	Hjalmar
22/5 supervision 5: Text classification (continued) and Quantitative (text) analysis		1) Work on quantitative text analysis 2) Label test set for evaluating classifier	Hjalmar

27/5 Lecture 8: Mixed Method analysis design and reporting	TBA	1) Clarify and justify mixed method analysis design 2) Continue with groups work	Hjalmar
30/5 Supervision 6:	**Milestone 4** 30 min per group depending on the number of groups and size of groups. A slideshow with 7 slides: 1) Research question and motivation 2) Description of datasite and topic 3) Data design 4) 3 slides for analysis and results(BOTH qual and quant) 5) Conclusion 30 min per group depending on the number of groups and size of groups. Half for presentation and half for peer and TA feedback. Upload presentation with group name and class.	Groups present their preliminary final analyses for each other – with peer feedback + feedback from TAs	
EXAM	Exam: hand-in 12/6; oral exams week 25 Full readings: approximately 400 pages + 80 pages self-selected literature.		Hjalmar